

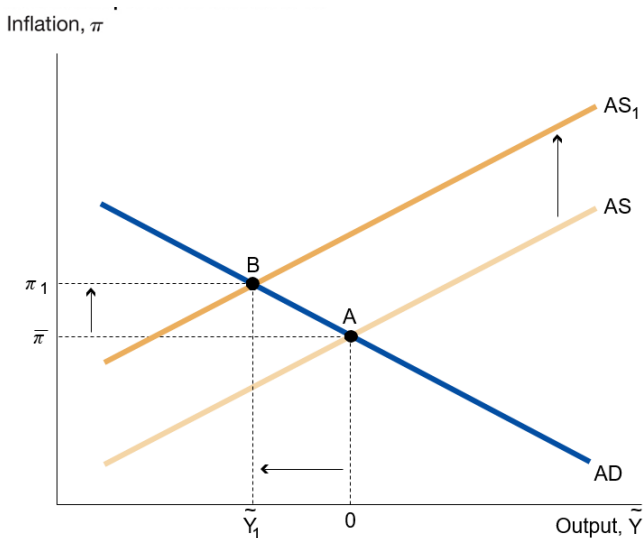
ECON 311

Lecture 18: AS-AD Model, part II

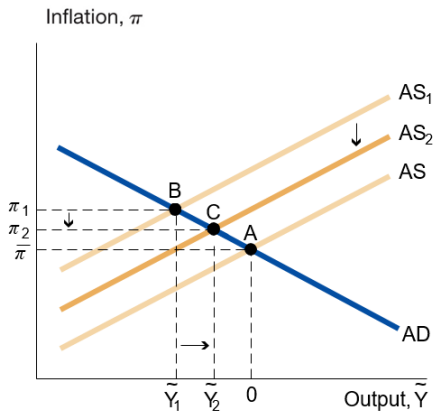
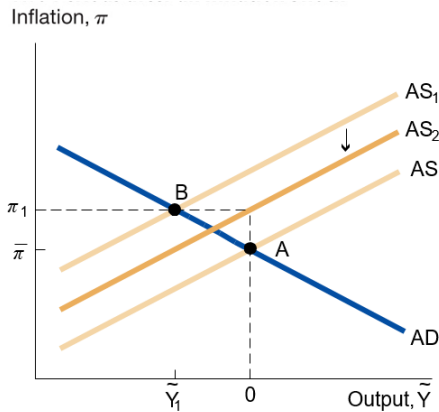
Using the AS-AD Model

Financial Crisis of 2008

Cost-push Shock

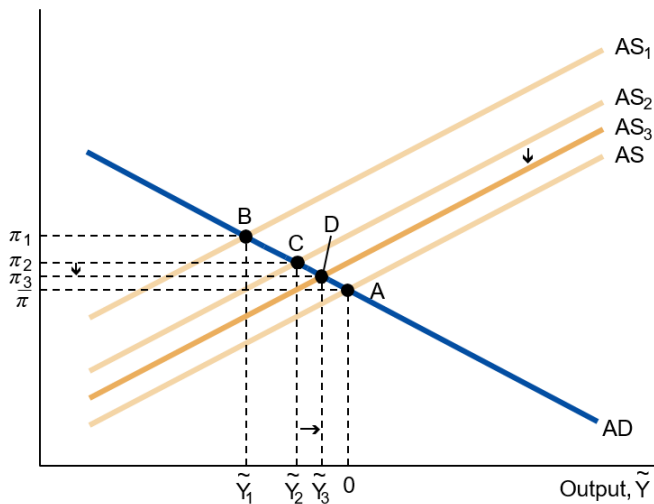


Inflation Shock

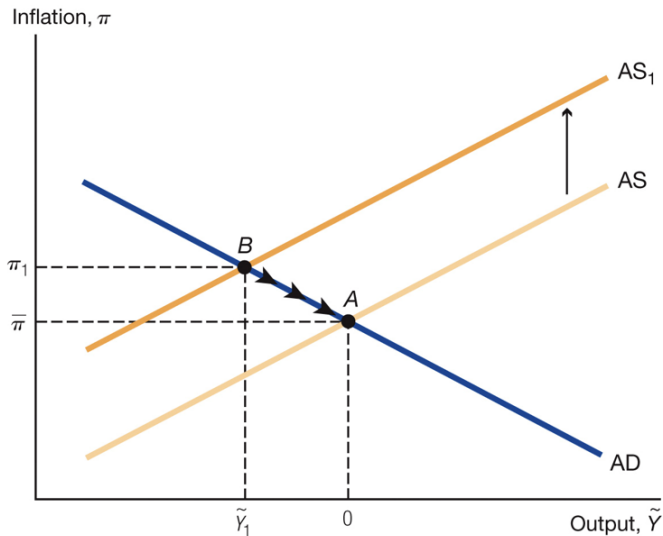


Cost-push Shock

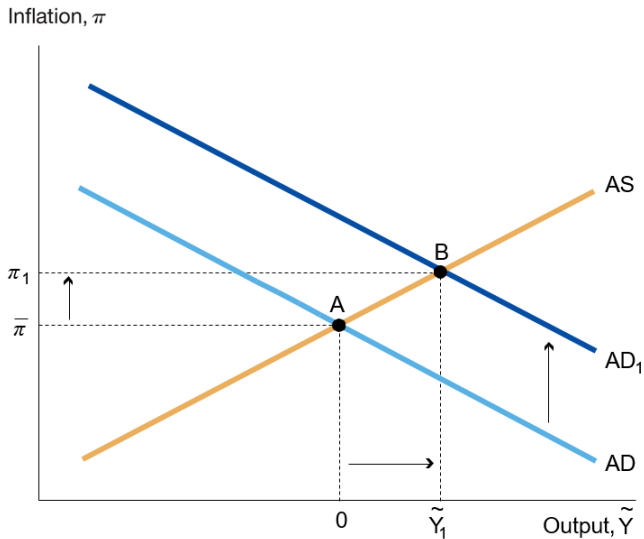
Inflation, π



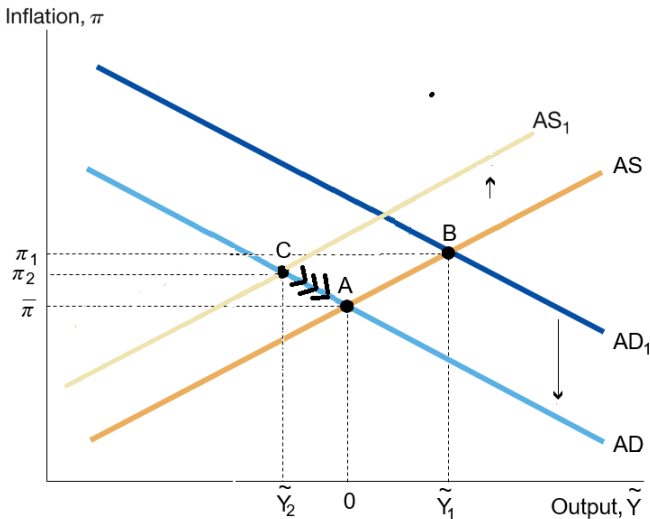
Cost-push Shock



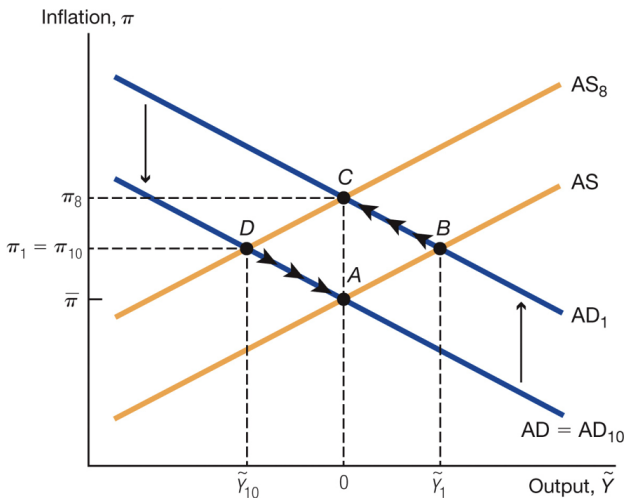
One Period Demand Shock



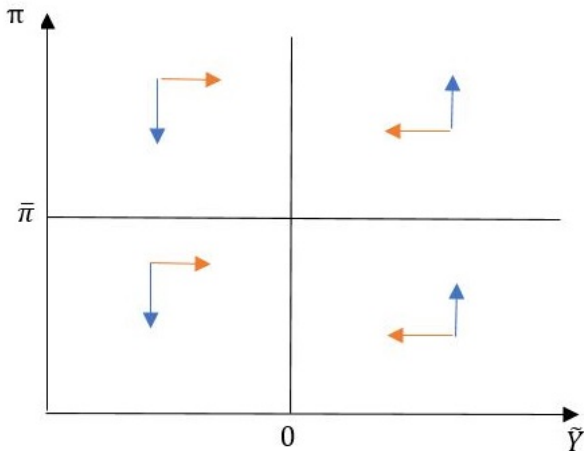
One Period Demand Shock



Multiple Period Demand Shock

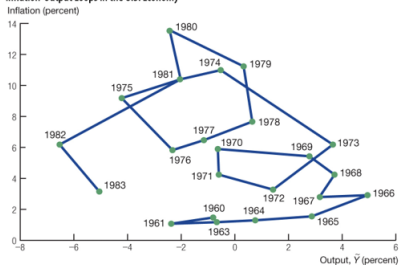


Phase diagram cost push and demand shocks

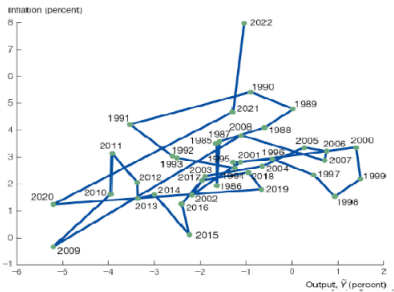


Empirical Evidence

Inflation-Output Loops in the U.S. Economy



(a)



(b)

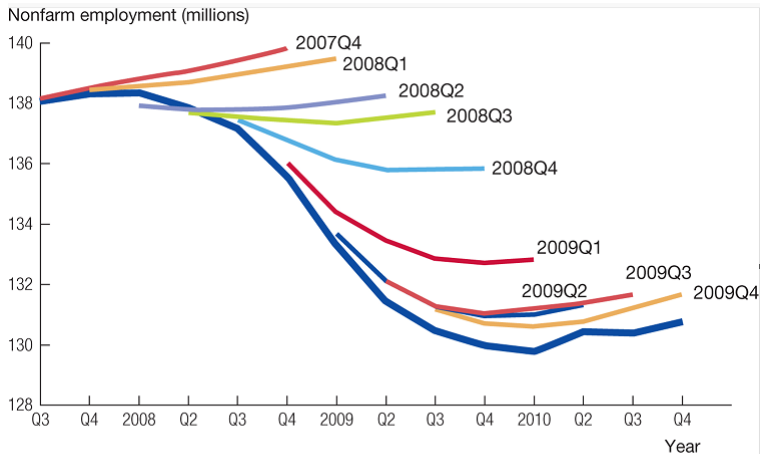
Rules vs. Discretion

- Explicit Policy Rules:
 - ▶ (pros) Help anchor inflation expectations and maintain a stable inflation environment where shocks are deflected quickly
 - ▶ (cons) To work, policy rules must be credible, and, therefore, provide little flexibility to deviate from targets
- Discretionary Monetary Policy:
 - ▶ (pros) Provides flexibility to pursue expansionary economic policy
 - ▶ (cons) Shocks can be very disruptive as inflation pressures do not dissipate quickly
- It is extremely important that, in Modern Economies, CBs are independent from the government in power

Using the AS-AD Model

Financial Crisis of 2008

Great Recession



Source: James H. Stock and Mark W. Watson, "Twenty Years of Time Series Econometrics in Ten Pictures," *Journal of Economic Perspectives* (Spring 2017).

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Great Recession: Shocks

Sequence of events that led to the crisis

- Global Savings Glut
 - ▶ In 1996, developing countries borrowed from the rest of the world.
 - ▶ In 2003, developing countries were saving a net \$205 billion into the world's capital markets.
 - ▶ Most of these savings enter the US market
- Housing Market and Subprime lending
- Rise interest rates and decrease housing prices
- Oil Prices

Great Recession: Shocks

Sequence of events that led to the crisis

- Global Savings Glut
- Housing Market and Subprime lending
 - ▶ Free Money led to subprime lending in housing markets through a variety of financial instruments deemed as "safe" assets (securitization)
 - ▶ The system worked as long as housing prices were increasing
- Rise interest rates and decrease housing prices
- Oil Prices

Great Recession: Shocks

Sequence of events that led to the crisis

- Global Savings Glut
- Housing Market and Subprime lending
- Rise interest rates and decrease housing prices
 - ▶ Between 2004 and 2006, the Fed raised its interest rate from 1.25% to 5.25% $\Rightarrow$ Higher interest rates ($>$ teaser) for subprime debtors
 - ▶ From mid-2006 to the beginning of 2012, Housing prices plummeted by 36% (bubble burst)
- Oil Prices

Great Recession: Shocks

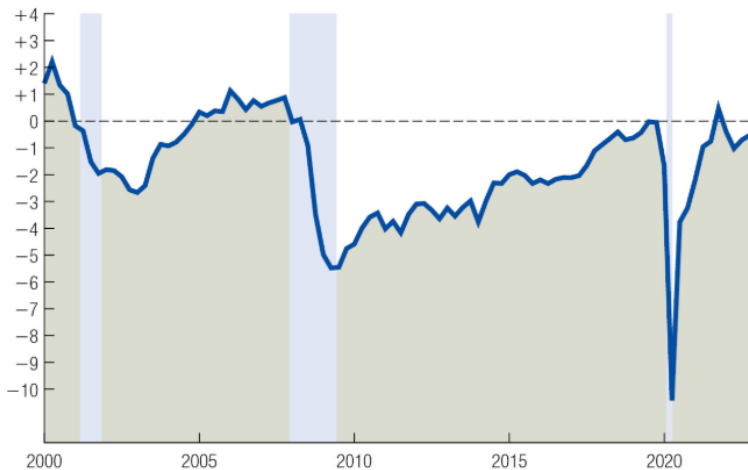
Sequence of events that led to the crisis

- Global Savings Glut
- Housing Market and Subprime lending
- Rise interest rates and decrease housing prices
- Oil Prices
 - ▶ In 2002, oil was about \$20 per barrel.
 - ▶ During the summer of 2008, prices peaked at \$140 per barrel.

Great Recession

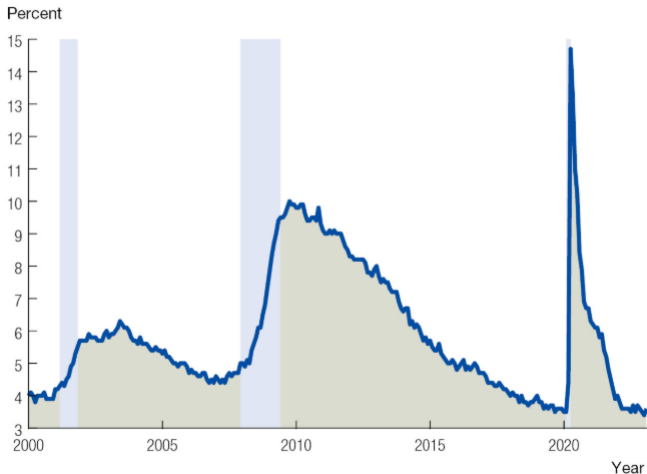
U.S. Short-Run Output, $\tilde{Y}$

Short-run output, $\tilde{Y}$ (percent)



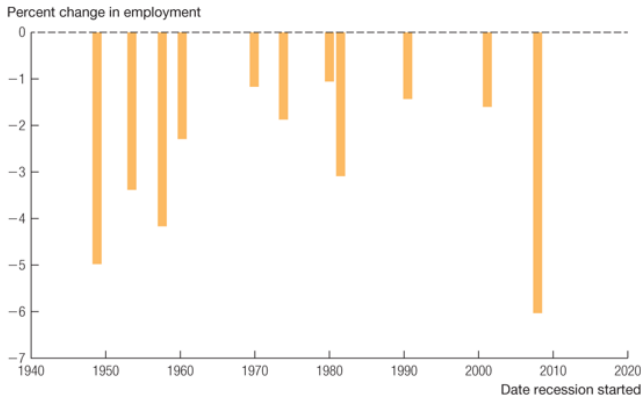
Great Recession

The U.S. Unemployment Rate



Source: The FRED database.

Great Recession

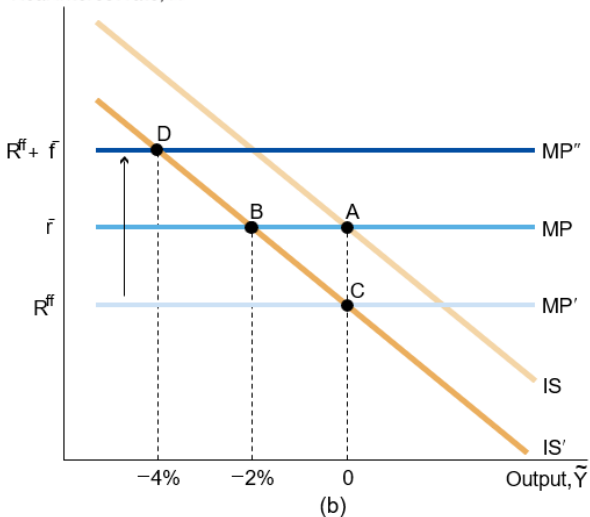


Source: U.S. Bureau of Labor Statistics, Civilian Unemployment Rate [UNRATE], retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/UNRATE>, September 22, 2019.
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Great Recession

A Housing Bubble and a Financial Crisis

Real interest rate, R



Great Recession

The Financial Crisis in the AS/AD Framework
Inflation, π

